# JINCHAO SONG

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## EDUCATION

2020 Ph.D. Department of Geosciences and Natural Resource Management, University of Copenhagen
2015 M.Sc. Chinese Academy of Sciences
2012 B.A.Sc. Ludong University, China

## ACADEMIC EXPERIENCES

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|---|---|
| 2024 – Current | Postdoctoral research fellow, University of Michigan |
| 2021 – 2024 | Postdoc, University of Southern Denmark |
| 2020 – 2021 | Postdoctoral Researcher, Institute of GIS and Remote Sensing, Peking University. |
| 2015 – 2016 | Research Assistant, Institute of Urban Environment, Chinese Academy of Sciences. |

## SELECTED RESEARCH HIGHLIGHT

**Nature Cities (2026) | First author**

*Weighing 600 million buildings reveals global urban material inequality and efficiency paths.*

Global mapping of building material stocks across more than 600 million buildings, revealing how urban form shapes material inequality and resource efficiency. The study was accompanied by a *Nature Cities Research Briefing* featuring editorial commentary by Senior Editor Wei Chen and expert opinion by Ruidong Chang (Adelaide University).

## PEER-REVIEWED JOURNAL ARTICLES

**Song, J.,** Sun, K., Goldstein, B., Tong, X., Wang, L., Helbich, M., Liu, Q., Dai, M., Li, X., & Liu, G. (2026). Weighing 600 million buildings reveals global urban material inequality and efficiency paths. ***Nature Cities***. https://doi.org/10.1038/s44284-026-00510-3

Liu, Q., Ouyang, X., Zhu, W., Sun, K., **Song, J.,** Li, X., Li, Y., Chen, W., & Liu, G. (2025). Spatiotemporal Characteristics of Global Building Material Intensity Revealed for Circular and Low-Carbon Construction. **Environmental Science & Technology.**

Sun, K., Li, Q., Dai, M., **Song J**., Liu G. *et al.* A global dataset for steel aluminum and cement in-use stocks at 500 m gridded level 2000-2019. ***Sci Data*** **12**, 1560 (2025).

Sun K, Li Q, Liu Q, **Song J,** Dai M, Qian X, Gummidi SR, Yu B, Creutzig F, Liu G. Urban Fabric Decoded: High-Precision Building Material Identification via Deep Learning and Remote Sensing. **Environmental Science and Ecotechnology**. 2025 Feb 3:100538.

**Song, J.,** Chen, W., Zhang, J., Huang, K., Hou, B., & Prishchepov, A. V. (2020). Effects of building density on land surface temperature in China: Spatial patterns and determinants. **Landscape and Urban Planning,** 198, 103794.

**Song, J.,** Zhao, C., Lin, T., Li, X., & Prishchepov, A. V. (2019). Spatio-temporal patterns of traffic-related air pollutant emissions in different urban functional zones estimated by real-time video and deep learning technique. **Journal of Cleaner Production**, 117881.

**Song, J.,** Zhao, C., Zhong, S., Nielsen, T. A. S., & Prishchepov, A. V. (2019). Mapping spatio-temporal patterns and detecting the factors of traffic congestion with multi-source data fusion and mining techniques. **Computers, Environment and Urban Systems**, *77*, 101364.

**Song, J.,** Tong, X., Wang, L., Zhao, C., & Prishchepov, A. V. (2019). Monitoring finer-scale population density in urban functional zones: A remote sensing data fusion approach. **Landscape and Urban Planning**, 103580.

Tang, C., Yang, C., Cai, R. S., Ye, H., Duan, L., Zhang, Z., Shi, Z., Lin, K., **Song, J.,** Huang, X., Zhang, H., Yang, J.,

& Cai, P. (2019). Analysis of the relationship between electromagnetic radiation characteristics and urban functions in highly populated urban areas. **Science of the Total Environment**, 654, 535-540.

**Song, J.,** Lin, T., Li, X., & Prishchepov, A. (2018). Mapping Urban Functional Zones by Integrating Very High Spatial Resolution Remote Sensing Imagery and Points of Interest: A Case Study of Xiamen, China. **Remote Sensing**, *10*(11), 1737.

T. Lin, X. Liu, **J. Song** et al., "Urban waterlogging risk assessment based on internet open data: A case study in China," **Habitat Int.,** vol. 71, pp. 88–96, 2018.

H. Ye, Q. Ren, L. Shi, **J. Song** et al., "The role of climate, construction quality, microclimate, and socio-economic conditions on carbon emissions from office buildings in China," **J. Clean. Prod**., vol. 171, 2018.

X. Li, **J. Song**, T. Lin, J. Dixon, G. Zhang, and H. Ye, "Urbanization and health in China, thinking at the national, local and individual levels," **Environ. Heal.** A Glob. Access Sci. Source, vol. 15, 2016.

**Song, J.** C., Li, X. H., Lin, T., Zhang, G. Q., Ye, H., He, X. Y., & Ge, R. B. (2015). A method of extracting urban built-up area based on DMSP/OLS nighttime data and Google Earth. *Journal of Geo-Information Science*, *17*(6), 750-756.

## RESEARCH BRIEFINGS

Liu, G., & **Song, J.** (2026). Global characterization of building-level weight reveals urban material inequality. Nature Cities. Research Briefing. https://doi.org/10.1038/s44284-026-00509-w.

## TEACHING AND MENTORING

**Remote Sensing** (EAS 541.001), University of Michigan. Sensing the city: Remote Sensing and Geo-Big Data in Urban Environmental Studies. Delivered one guest lecture as part of the course.

**Sustainable Urban Systems** (EAS 501.074), University of Michigan (Winter 2026). One guest lecture

**GIS for Engineering Sustainability,** University of Southern Denmark (F22) (February 2022 - June 2022 and 2023). Lecturer (master's, 5 ECTS, English): course planning, lectures, practicals and project supervision.

**Remote sensing in land use (March 2018 and March 2017).** Geospatial monitoring and modelling using TerrSet.

**ArcGIS (October 2015)** Responsibility: Lecture on geospatial statistics based on ArcGIS software

**Master's research mentoring (graduates): Kunyu Yang** (University of Michigan): technical and manuscript writing guidance throughout the project; corresponding author of her manuscript. **Mias Sommer Schjønberg** (University of Southern Denmark): day-to-day research guidance.

## MEDIA AND RESEARCH HIGHLIGHTS

2026 **University of Michigan SEAS** (17 September). "Weighing 600 million buildings to help construct a more efficient future".

2026 **Anthropocene** (15 September). "Why less steel and concrete isn't the same as more sustainable".

2026 **Michigan News** (14 September). Weighing 600 million buildings to help construct a more efficient future.

2026 **EurekAlert!** (11 September). Weighing 600 million buildings to help construct a more efficient future.

2026 **Tech Xplore** (11 September). https://techxplore.com/news/2026-09-million-efficient-future.html

2026 **Peking University** (10 September). Research news on global building material stocks and sustainable urban transitions (in Chinese).

2026 **The Sun**. "Weight of the world". (Print edition, 2026).

## MANUSCRIPTS UNDER REVIEW

2026 Goldstein B., **Song J.,** Gummidi B., Elliot T. Spatial tipping points in the metabolism of cities. **Nature Cities** (Under Review)

2026 **Song J.,** Du S., Du D, Yang K, Goldstein B., Multi-scale analysis of seasonal thermal dynamics and landscape thresholds across Chinese cities. **Urban Climate** (Under Review)

2026 Yang K, **Song J.,** Goldstein B. Mapping resort-driven land transformation and forest carbon loss in the Yucatán Peninsula, Mexico. **Journal of Environmental Management** (Under Review)

2026 **Song J.,** Helbich M., Du D., Hou B., Liu J., & Goldstein B. Monitoring urban traffic emissions using high-resolution satellite imagery and near-real-time congestion data. **Sustainable Cities and Society** (Under Review)

## CONFERENCE PRESENTATIONS

2025 AAG Annual Meeting. Weighing the world's over 600 million buildings reveals stark global disparities and impending material demands in the Global South. **Jinchao Song***

2018 International Geographical Union, IGU thematic conference, Moscow. Impact of Functional Zones on Traffic Congestion Using Real–Time Traffic Data. **Jinchao Song***

2018 International Geographical Union, IGU thematic conference, Moscow. Modelling Urban Population Distribution Using Multi–Source Remote Sensing Imagery and Location–Based Service Networks. **Jinchao Song***

## CONFERENCE POSTERS

2026 Gordon Research Conference on Industrial Ecology, Maine, USA. Weighing 600 million buildings reveals global urban material inequality and efficiency paths. **Jinchao Song**

2017 American Geophysical Union Fall Meeting, New Orleans. Simulating population density using high resolution imagery and location-based service data. **Jinchao Song**

## JOURNAL REVIEWER

Remote Sensing of Environment; Cities; Scientific Data

## EDITOR ACTIVITIES

2024 **Guest-Editor**, Special Issue on Remote Sensing: Remote Sensing and GIS for Monitoring Urbanization and Urban Health, Xinhu Li, Shihong Du, Peng Jia, Franz W. Gatzweiler, **Jinchao Song**

2025 **Guest-Editor**, Special Issue on Remote Sensing: Remote Sensing and GIS for Monitoring Urbanization and Urban Health (Second Edition), Xinhu Li, Shihong Du, Peng Jia, Franz W. Gatzweiler, **Jinchao Song**

2025 **Guest-Editor**, Special Issue on Applied Sciences: Intelligent Understanding and Analysis of Urban Big Data, Shouhang Du, **Jinchao Song**, Xiuyuan Zhang

## FELLOWSHIPS

2020 Boya Postdoctoral Fellowship, Peking University, China (88,000 Euro)

## RESEARCH GRANTS AND FUNDING

2021 Postdoctoral research funding, China Postdoctoral Science Foundation ($11,000 USD)

## SCHOLARSHIPS

2015 Scholarship from Institute of Urban Environment, Chinese Academy of Sciences (1,300 Euro)

2014 Scholarship from Institute of Urban Environment, Chinese Academy of Sciences (1,300 Euro)

## GEOSPATIAL METHODS AND TECHNICAL SKILLS

**GIS and geospatial computing:** raster and vector geoprocessing; spatial tiling; multi-source geospatial data integration; Python; ArcGIS.

**Remote sensing and image analysis:** Landsat/Sentinel-2 image filtering, cloud masking and monthly compositing; land-cover classification; urban functional zone mapping; building footprint extraction and height estimation; Google Earth Engine (GEE); ENVI.

**Urban climatology and thermal remote sensing:** land surface temperature retrieval; surface urban heat island assessment; seasonal thermal hysteresis analysis; surface energy-balance modeling.

**Geospatial machine learning:** random forest and XGBoost modeling; k-means clustering; partial dependence analysis; deep-learning object detection and counting in satellite imagery and video (YOLO, Mask R-CNN).

**Transport geography and emissions modeling:** spatiotemporal analysis of traffic congestion; road network and urban functional zone analysis; traffic emissions estimation from vehicle counts, speeds and emission factors.

**Urban morphology and material stock modeling:** building-scale material stock estimation; urban form analysis; model validation and uncertainty assessment.

**Spatial statistics and geovisualization:** GeoDetector; spatial statistical analysis; thematic mapping and scientific visualization.